

JASMEHAR KAUR

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Education

University of Waterloo

Honours Software Engineering

Sep. 2024 – Present

Waterloo, ON

Technical Skills

Languages: Python, Java, C++, C, JavaScript, TypeScript, HTML/CSS, SQL

Frameworks/Libraries: PyTorch, Scikit-learn, LangChain, Flask, FastAPI, Docker, Kubernetes, Helm, Node.js, React.js, React Native, TailwindCSS, Tesseract OCR

Tools/Technologies: Git, GitHub, Bash, MATLAB, Postman, AWS (Athena, Lambda, API Gateway), WebSocket, Raspberry Pi, Linux, REST APIs, Agile, RAG, OpenSearch, Selenium

Experience

Nokia

Apr. 2025 – Aug. 2025

AI/ML and Full-Stack Developer

Ottawa, ON

- Built a cloud-native **multi-agent LLM**-based system to automate 5G alarm resolution, with **vector-based RAG**-driven chain-of-thought analysis for high-relevance retrieval.
- Engineered a **RAG** pipeline by optimizing **hybrid search** (dense + sparse) in **OpenSearch** and implementing multi-pass retrieval, achieving a **65%** increase in precision and **58%** increase in recall.
- Developed a **Crossplane**-based cloud automation tool and a **custom Kubernetes operator** in **Go** to manage the lifecycle of autonomous applications and enable intent-based network orchestration.
- Built interactive **React** frontends with modular components and dynamic theming.

Trexo Robotics (YC '19)

Jul. 2022 – Sep. 2022

Software Developer Intern

Mississauga, ON

- Developed a **serverless data management system** with React, Redux, and **AWS Athena** via **API Gateway**, enabling authenticated **CRUD operations** on large datasets and cutting data editing time by **92%**.
- Built **custom API endpoints** and scripts to automate manual data entry, establishing a **controlled, programmatic interface** for database interactions and reducing data entry errors.

Absolute Robotics

Sep. 2022 – Jul. 2024

IT Lead

Mississauga, ON

- Engineered a **data collection and analytics** system with a **React Native** app for real-time match data capture and a **React** web app for offline ingestion via QR codes, reducing data entry time by **95%**.

Projects

dill.pkl | FastAPI, React, TypeScript, Python, Optuna, OpenRouter

- Built an **8-agent AutoML** web app automating the full ML pipeline — data analysis, preprocessing, feature engineering, model selection, training, and evaluation — outputting a fully deployed, documented model.
- Designed a **dependency-aware rerun engine** and **controlled action registry** enabling natural-language pipeline modification, keeping LLM intent separate from execution for safe, deterministic changes.
- Parallelized training across candidate models with per-candidate **Optuna HPO**, packaging the best by CV score into a deployment zip with a **schema-validated FastAPI endpoint** and auto-generated README.

iClick | Scikit-learn, OpenCV, MediaPipe, NumPy, Python, pandas

- Built a real-time vision pipeline for **hand and eye tracking** with robust noise and drop-frame handling.
- Developed end-to-end calibration and training workflows, reducing gaze error to **8 px MAE** on held-out data.
- Engineered a multi-process vision, voice, and UI control stack, improving **long-session stability**.
- Added validation and robustness logic (**smoothing, clamping, cooldowns**), improving tracking consistency by **72%**.

FOOT.print | Gemini API, Python, OpenCV, YOLOv8, MiDaS, TwelveLabs, Blender

- Architected and deployed a full-stack AI pipeline converting raw room video into optimized 3D layouts.
- Built a **CV/NLP model** using **Gemini** and YOLOv8 to predict object sizes and generate accurate layout recommendations.
- Automated **Blender** mesh creation and validation loops using Gemini, producing geometrically accurate 3D models.

Breast Cancer Prediction Model | Python, PyTorch, Scikit-learn

- Developed and optimized a **feed-forward neural network** in **PyTorch** for binary classification, achieving **97.37% accuracy** on the test set after 100 epochs.
- Engineered an end-to-end ML pipeline with data preprocessing (standardization, train-test split) via **Scikit-learn**, trained with **Adam optimizer** and **Binary Cross-Entropy Loss** for robust generalization.

Certifications

AWS Cloud Technical Essentials | AWS on Coursera

- Built foundational knowledge in cloud computing principles and core AWS services.